14.4: Many human traits follow Mendelian patterns of inheritance

The study of human genetics is fueled by our desire to understand our own inheritance and to develop treatments for genetically based diseases. While peas are convenient subjects for genetic research, humans are not. The human generation span is long (about 20 years), human parents produce many fewer offspring than peas, and breeding experiments would be unethical!

Pedigree Analysis

In place of breeding experiments, geneticists analyze the results of human matings that have already occurred. They collect information about a family's history for a particular trait and assemble this information into a family tree describing the trait across the generations—a family **pedigree**.

Figure 14.15a shows a three-generation pedigree that traces the occurrence of a pointed contour of the hairline on the forehead. We simplify the genetics of this trait, called a widow's peak, by assuming it is due to a dominant allele, W. (In reality, other genes are probably also involved.) In our model, because the widow's peak allele is dominant, all individuals who lack a widow's peak must be homozygous recessive (ww). The two grandparents with widow's peaks must have the Ww genotype since some of their offspring are homozygous recessive. The offspring in the second generation who *do* have widow's peaks must also be heterozygous because they are the products of Ww×ww matings. The third generation in this pedigree consists of two sisters. The one who has a widow's peak could be either homozygous (WW) or heterozygous (Ww), given what we know about the genotypes of her parents (both Ww).

Figure 14.15

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Pedigree analysis.
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Pedigree showing three generations of a family with respect to widow's peak and ability to taste PTC. Images of two sisters shown in the third generation; one with widow's peak and one without.

Each pedigree traces a trait through three generations of the same family; the two traits have different inheritance patterns. For the sake of understanding genetic principles, each trait is shown as being determined by two alleles of a single gene—a simplification, since other genes likely also affect these two characters.

Part a.

- First generation (grandparents): Pedigree drawn with symbols as follows from left to right: First grandparents: Shaded Square (uppercase W lowercase w) with a line drawn to circle (w w, both lowercase). Space and then second grandparents: square (w w both lowercase) with a line drawn to shaded circle (uppercase W lower case w).
- Second generation (parents, uncles, and aunts): Children of first grandparents: Shaded circle (uppercase W lowercase w), square (w w both lowercase), square (w w both lowercase), shaded square (uppercase W lowercase w). This individual has a line drawn to the first child of the second grandparents, a shaded circle (uppercase W lowercase w). Next is a circle (w w both lowercase).
- Third generation: children of the couple in the second generation: Shaded circle (W W both uppercase or uppercase W lowercase w). This individual is shown next to a photograph of a woman with Widow's peak. Next is a circle (w w both lowercase). This individual is shown next to a photograph of a woman with no widow's peak.
- Is a widow's peak a dominant or recessive trait? Tips for pedigree analysis: Notice in the third generation that the second-born daughter lacks a widow's peak, although both of her parents had the trait. Such a pattern indicates that the trait is due to a dominant allele. If it were due to a recessive allele, and both parents had the recessive phenotype (straight hairline), all of their offspring would also have the recessive phenotype.

Part b.

- Pedigree tracing the ability to taste P T C through 3 generations of a family. In the third generation, photos show two sisters; one can taste PTC and the other can't.
- First generation (grandparents): Pedigree drawn with symbols as follows from left to right: first grandparents: Square (uppercase T lowercase t) with line drawn to circle (uppercase T lowercase t). Space then second set of grandparents: shaded square (tt both lowercase) with line drawn to circle (uppercase T lowercase t).
- Second generation (parents, aunts, uncles). Children of first set of grandparents: Circle (T T both uppercase or uppercase T lowercase t), shaded square (t t both lowercase), shaded square (t t both lowercase), square (uppercase T lowercase t). This individual has a line drawn to child of second grandparents: circle (uppercase T lowercase t) and the other child of the second grandparents is a shaded circle (t t both lowercase).
- Children of the second-generation couple are in the third generation (two sisters). First a shaded circle (t t both lowercase, next to a photo of one sister who is smiling because she cannot taste P T

C. Then a circle (TT both uppercase or uppercase T lowercase t, next to a photo of her sister who is grimacing because she can taste the bitter chemical P T C.

Figure 14.15b is a pedigree of the same family, but this time we focus on a recessive trait, the inability of individuals to taste a chemical called PTC (phenylthiocarbamide). Compounds similar to PTC are found in broccoli, brussels sprouts, and related vegetables and account for the bitter taste some people report when eating these foods. We'll use t for the recessive allele and T for the dominant allele, which results in the ability to taste PTC. As you work your way through the pedigree, notice once again that you can apply what you have learned about Mendelian inheritance to understand the genotypes shown for the family members.

An important application of a pedigree is to help us calculate the probability that a future child will have a particular genotype and phenotype. Suppose that the couple represented in the second generation of Figure 14.15 decides to have one more child. What is the probability that the child will have a widow's peak? This is equivalent to a Mendelian F1 monohybrid cross Ww×Ww, and therefore the probability that a child will inherit a dominant allele and have a widow's peak is 34(14WW+12Ww). What is the probability that the child will be unable to taste PTC? We can also treat this as a monohybrid cross Tt×Tt, but this time we want to know the chance that the offspring will be homozygous recessive (tt). That probability is 14. Finally, what is the chance that the child will have a widow's peak *and* be unable to taste PTC? Assuming that the genes for these two characters are on different chromosomes, the two pairs of alleles will assort independently in this dihybrid cross WwTt×WwTt. Therefore, we can use the multiplication rule:

34(chance of widow's peak)×14(chance of inability to taste PTC)=316
(chance of widow's peak and inability to taste PTC).

Pedigrees are a more serious matter when the alleles in question cause disabling or deadly diseases instead of innocuous human variations such as hairline or inability to taste a harmless chemical. However, for disorders inherited as simple Mendelian traits, the same techniques of pedigree analysis apply.

Recessively Inherited Disorders

Thousands of genetic disorders are known to be inherited as simple recessive traits. These disorders range in severity from relatively mild, such as albinism (lack of pigmentation, which results in susceptibility to skin cancers and vision problems), to life-threatening, such as cystic fibrosis.

The Behavior of Recessive Alleles

How can we account for the behavior of alleles that cause recessively inherited disorders? Recall that genes code for proteins of specific function. An allele that causes a genetic disorder (let's call it allele a) codes for either a malfunctioning protein or no protein at all. In the case of disorders classified as recessive,

heterozygotes (Aa) typically have the normal phenotype because one copy of the normal allele (A) produces a sufficient amount of the specific protein. Thus, a recessively inherited disorder shows up only in the homozygous individuals (aa) who inherit a recessive allele from each parent. Although phenotypically normal with regard to the disorder, heterozygotes may transmit the recessive allele to their offspring and thus are called **carriers**. **Figure 14.16** illustrates these ideas using albinism as an example.

Figure 14.16

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Albinism: a recessive trait.
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Image of two sisters, one with albinism and one with normal coloration. Punnett square showing the results of cross between parents with normal phenotypes who are both carriers for albinism.

One of the two sisters shown here does not have albinism; the other does. Most recessive homozygotes are born to parents who are carriers of the disorder but themselves have a normal phenotype, the case shown in the Punnett square.

Parents both normal phenotypes, and both with genotypes uppercase A lowercase a.

Probabilities for phenotypes in their children are one-fourth normal, one-half normal carriers, and one-fourth albino.

Gametes	Uppercase A	Lowercase a
Uppercase A	A A both uppercase Phenotype: normal	Uppercase A lower case a Phenotype: Normal (carrier)
Lowercase a	Uppercase A lower case a Phenotype: Normal (carrier)	a a both lowercase Phenotype: albino

What is the probability that the sister without albinism is a carrier of the albinism allele?

*For suggested answer, see*Appendix A.**

Most people who have recessive disorders are born to parents who are carriers of the disorder but have a normal phenotype, as is the case shown in the Punnett square in Figure 14.16. A mating between two carriers corresponds to a Mendelian F1 monohybrid cross, so the predicted genotypic ratio for the offspring is 1AA:2 Aa:1aa. Thus, each child has a 14 chance of inheriting a double dose of the recessive allele; in the case of albinism, such a child will have albinism. From the genotypic ratio, we also can see that out of three offspring with the *normal* phenotype (one AA plus two Aa), two are predicted to be heterozygous carriers, a

23 chance. Recessive homozygotes could also result from Aa×aa and aa×aa matings, but if the disorder is lethal before reproductive age or results in sterility (neither of which is true for albinism), no aa individuals will reproduce. Even if recessive homozygotes are able to reproduce, this will occur relatively rarely because such individuals are much less common in the population than heterozygous carriers (for reasons we'll examine in Concept 23.2).

In general, genetic disorders are not evenly distributed among all groups of people. For example, the incidence of Tay-Sachs disease, which we described earlier in this chapter, is disproportionately high among Ashkenazic Jews, Jewish people whose ancestors lived in central Europe. In that population, Tay-Sachs disease occurs in one out of 3,600 births, an incidence about 100 times greater than that among non-Jews or Mediterranean (Sephardic) Jews. This uneven distribution results from the different genetic histories of the world's peoples during less technological times, when populations were more geographically (and hence genetically) isolated.

When a disease-causing recessive allele is rare, it is relatively unlikely that two carriers of the same harmful allele will meet and mate. The probability of passing on recessive traits increases greatly, however, if the man and woman are close relatives (for example, siblings or first cousins). This is because people with recent common ancestors are more likely to carry the same recessive alleles than are unrelated people. Thus, these consanguineous ("same blood") matings, indicated in pedigrees by double lines, are more likely to produce offspring homozygous for recessive traits—including harmful ones. Such effects can be observed in many types of domesticated and zoo animals that have become inbred.

Although geneticists generally agree that inbreeding causes an increase in autosomal recessive conditions compared to those resulting from matings between unrelated parents, they debate exactly how much human consanguinity increases the risk of inherited diseases. For one thing, many harmful alleles have such severe effects that a homozygous embryo spontaneously aborts long before birth. Most societies and cultures have laws or taboos forbidding marriages between close relatives, some for social or economic reasons. These rules may have also resulted from the empirical observation that in most populations, stillbirths and physical or biochemical abnormalities at birth are more common when parents are closely related.

Cystic Fibrosis

The most common lethal genetic disease in the United States is **cystic fibrosis**, which strikes one out of every 2,500 people of European descent but is much rarer in other groups. Among people of European descent, one out of 25 4% are carriers of the cystic fibrosis allele. The normal allele for this gene codes for a membrane protein that functions in the transport of chloride ions between certain cells and the extracellular fluid. These chloride transport channels are defective or absent in the plasma membranes of children who inherit two recessive alleles for cystic fibrosis. The result is an abnormally high concentration of intracellular

chloride, which causes an uptake of water due to osmosis. This in turn causes the mucus that coats certain cells to become thicker and stickier than normal and to build up in the pancreas, lungs, digestive tract, and other organs. Multiple (pleiotropic) effects result, including poor absorption of nutrients from the intestines, chronic bronchitis, and recurrent bacterial infections.

Untreated, cystic fibrosis can cause death by the age of 5. Daily doses of antibiotics to stop infection, gentle pounding on the chest to clear mucus from clogged airways, and other therapies can prolong life. In the United States, more than half of those with cystic fibrosis now survive into their 40s.

Sickle-Cell Disease: A Genetic Disorder with Evolutionary Implications

EVOLUTION The most common inherited disorder among people of African descent is **sickle-cell disease**, which affects one out of 400 African-Americans. Sickle-cell disease is caused by the substitution of a single amino acid in the hemoglobin protein of red blood cells; in homozygous individuals, all hemoglobin is of the sickle-cell (abnormal) variety. When the oxygen content of an affected individual's blood is low (at high altitudes or under physical stress, for instance), the sickle-cell hemoglobin proteins aggregate into long fibers that deform the red cells into a sickle shape (see Figure 5.19). Sickled cells may clump and clog small blood vessels, often leading to other symptoms throughout the body, including physical weakness, pain, organ damage, and even stroke and paralysis. Regular blood transfusions can ward off brain damage in children with sickle-cell disease, and new drugs can help prevent or treat other problems. There is currently no widely available cure, but the disease is the target of ongoing gene therapy research.

Although two sickle-cell alleles are necessary for an individual to have sickle-cell disease and thus the condition is considered recessive, the presence of one sickle-cell allele can affect the phenotype. Thus, at the organismal level, the normal allele is incompletely dominant to the sickle-cell allele **(Figure 14.17)**. At the molecular level, the two alleles are codominant; both normal and abnormal (sickle-cell) hemoglobins are made in heterozygotes (carriers), who are said to have *sickle-cell trait*. (Here the word "trait" is used to distinguish this condition from sickle-cell disease, not as it was defined earlier—as any variant of a character.) Heterozygotes are usually healthy but may suffer some symptoms during long periods of reduced blood oxygen.

Figure 14.17

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Sickle-cell disease and sickle-cell trait.
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Sickle-cell disease and sickle-cell trait.

Part a. Homozygote with sickle-cell disease. Weakness, anemia, pain and fever, organ damage. Sickle cell alleles on two homologous chromosomes flows to sickle-cell hemoglobin proteins which goes with low O_2 to part of a fiber of sickle-cell hemoglobin proteins. Long fibers cause red blood cells to be sickle-shaped, leading to sickle cell disease.

Part b. Heterozygote with sickle-cell trait. Some symptoms when blood oxygen is very low; reduction of malaria symptoms. Pair of homologous chromosomes, one with a sickle-cell allele and one with a normal allele, flows to both sickle cell and normal hemoglobin proteins, flows with low O_2 to part of a sickle-cell fiber and normal hemoglobin proteins, which flows to both sickled and normal red blood cells, which flows to sickle-cell trait.

HHMI BioInteractive Animation

Sickle-Cell Disease

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Considering the severe detrimental effects of this allele in homozygotes, there is an unusually high percentage of heterozygotes. For example, about one out of ten African-Americans have sickle-cell trait. Why haven't evolutionary processes resulted in the disappearance of the allele among this population? One explanation is that having a single copy of the sickle-cell allele reduces the frequency and severity of malaria attacks, especially among young children. The malaria parasite spends part of its life cycle in red blood cells (see Figure 28.18), and the presence of even heterozygous amounts of sickle-cell hemoglobin results in lower parasite densities and hence reduced malaria symptoms. Thus, in tropical Africa, where infection with the malaria parasite is common, the sickle-cell allele confers an advantage to heterozygotes even though it is harmful in the homozygous state. (The balance between these two effects will be discussed in Concept 23.4; see Make Connections Figure 23.18.) The relatively high frequency of African-Americans with sickle-cell trait is a vestige of their tropical African ancestry.

HHMI BioInteractive Video

The Making of the Fittest: Natural Selection in Humans (Sickle-Cell Disease)

image

Play

image

Play

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Dominantly Inherited Disorders

Although many harmful alleles are recessive, some disorders are due to dominant alleles. One example is *achondroplasia*, a form of dwarfism that occurs in one of every 25,000 people. Heterozygous individuals have the dwarf phenotype **(Figure 14.18)**. Therefore, all people who do not have achondroplasia—99.99% of the population—are homozygous recessive. Like the presence of extra fingers or toes mentioned earlier, achondroplasia is a trait for which the recessive allele is much more prevalent than the corresponding dominant allele.

Figure 14.18

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Achondroplasia: a dominant trait.
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Punnett square shows the outcome of cross between a heterozygous dwarf and a normal (homozygous recessive) parent. Half their children will be heterozygous (dwarves) and half are normal (homozygous recessive).

See table next tab

Dr. Michael C. Ain has achondroplasia, a form of dwarfism caused by a dominant allele. This has inspired his work: He is a specialist in the repair of bone defects caused by achondroplasia and other disorders. The dominant allele (D) might have arisen as a mutation in the egg or sperm of a parent or could have been inherited from an affected parent, as shown for an affected father in the Punnett square.

Unlike achondroplasia, which is relatively harmless, some dominant alleles cause lethal diseases. Those that do are much less common than recessive alleles that have lethal effects. A lethal recessive allele is only lethal when homozygous; it can be passed from one generation to the next by heterozygous carriers because the carriers themselves have normal phenotypes. A lethal dominant allele, however, often causes the death of afflicted individuals before they can mature and reproduce, and in this case the allele is not passed on to future generations.

A lethal dominant allele may be passed on, though, if the lethal disease symptoms first appear after reproductive age. In these cases, the individual may already have transmitted the allele to his or her children. For example, a degenerative disease of the nervous system, called **Huntington's disease**, is caused by a lethal dominant allele that has no obvious phenotypic effect until the individual is about 35 to 45 years old. Once the deterioration of the nervous system begins, it is irreversible and inevitably fatal. As with other dominant traits, a child born to a parent with the Huntington's disease allele has a 50% chance of inheriting the allele and the disorder (see the Punnett square in Figure 14.18). In the United States, this disease afflicts about one in 10,000 people.

At one time, the onset of symptoms was the only way to know if a person had inherited the Huntington's allele, but this is no longer the case. By analyzing DNA samples from a large family with a high incidence of the disorder, geneticists tracked the Huntington's allele to a locus near the tip of chromosome 4, and the gene was sequenced in 1993. This information led to the development of a test that could detect the presence of the Huntington's allele in an individual's genome. (The methods that make such tests possible are discussed in Concepts 20.1 and 20.4.) The availability of this test poses an agonizing dilemma for those with a family history of Huntington's disease. Some individuals may want to be tested for this disease, whereas others may decide it would be too stressful to find out.

Interview

Interview with Nancy Wexler: Mapping the gene that causes Huntington's disease

Nancy Wexler.

Multifactorial Disorders

The hereditary diseases we have discussed so far are sometimes described as simple Mendelian disorders because they result from abnormality of one or both alleles at a single genetic locus. Many more people are susceptible to diseases that have a multifactorial basis—a genetic component plus a significant environmental influence. Heart disease, diabetes, cancer, alcoholism, certain mental illnesses such as schizophrenia and bipolar disorder, and many other diseases are multifactorial. In these cases, the hereditary component is polygenic. For example, many genes affect cardiovascular health, making some of us more prone than others to heart attacks and strokes. No matter what our genotype, however, our lifestyle has a tremendous effect on phenotype for cardiovascular health and other multifactorial characters. Exercise, a healthful diet, abstinence from smoking, and an ability to handle stressful situations all reduce our risk of heart disease and some types of cancer.

Genetic Testing and Counseling

Avoiding simple Mendelian disorders is possible when the risk of a particular genetic disorder can be assessed before a child is conceived or during the early stages of the pregnancy. Many hospitals have genetic counselors who can provide information to prospective parents concerned about a family history for a specific disease. Fetal and newborn testing can also reveal genetic disorders.

Counseling Based on Mendelian Genetics and Probability Rules

Consider the case of a hypothetical couple, Tyler and Lily. Each had a brother who died from the same recessively inherited lethal disease. Before conceiving their first child, Tyler and Lily seek genetic counseling to determine the risk of having a child with the disease. From the information about their brothers, we know that both parents of Tyler and both parents of Lily must have been carriers of the recessive allele. Thus, Tyler and Lily are both products of Aa×Aa crosses, where a symbolizes the allele that causes this particular disease. We also know that Tyler and Lily are not homozygous recessive (aa) because they do not have the disease. Therefore, their genotypes are either AA or Aa.

Given a genotypic ratio of 1AA:2Aa:1aa for offspring of an Aa×Aa cross, Tyler and Lily each have a 23 chance of being carriers (Aa). According to the rule of multiplication, the overall probability of their firstborn having the disorder is 23 (the chance that Tyler is a carrier) times 23 (the chance that Lily is a carrier) times 14 (the chance of two carriers having a child with the disease), which equals 19. Suppose that Lily and Tyler decide to have a child—after all, there is an 89 chance that their baby will not have the disorder. If, despite these odds, their child is born with the disease, then we would know that *both* Tyler and Lily are, in fact, carriers (Aa genotype). If both Tyler and Lily are carriers, there is a 14 chance that any subsequent child this couple has will have the disease. The probability is higher for subsequent children because the diagnosis of the disease in the first child established that both parents are carriers, not because the genotype of the first child affects in any way that of future children.

When we use Mendel's laws to predict possible outcomes of matings, it is important to remember that each child represents an independent event in the sense that its genotype is unaffected by the genotypes of older siblings. Suppose that Tyler and Lily have three more children, and *all three* have the hypothetical hereditary disease. There is only one chance in 64(14×14×14) that such an outcome will occur. Despite these circumstances, the chance that a fourth child of this couple will have the disease remains 14.

Tests for Identifying Carriers

Most children with recessive disorders are born to parents with normal phenotypes. The key to accurately assessing the genetic risk for a particular disease is therefore to find out whether the prospective parents are

heterozygous carriers of the recessive allele. Tests are available that can distinguish individuals of normal phenotype who are dominant homozygotes from those who are heterozygous carriers, using blood or cells from the inside of the cheek. Hundreds of different recessive alleles can be detected, including those for Tay-Sachs disease, sickle-cell disease, and cystic fibrosis.

These tests for identifying carriers enable people with family histories of genetic disorders to make informed decisions about having children, including whether to do genetic testing of the fetus, should they decide to become pregnant. The results can also lead carriers to alert family members that they may also be carriers and might choose to be tested. The tests raise other issues as well: Could carriers be denied health or life insurance or lose the jobs providing those benefits, even though they themselves are healthy? The Genetic Information Nondiscrimination Act, in place since 2008, allays these concerns by prohibiting discrimination in employment or health insurance coverage based on genetic test results. This is an important act that protects patients' rights but is unknown to many, including physicians. Genetic counselors, trained in a growing number of professional programs, can help individuals understand their genetic test results. Once test results are clearly understood, those with certain results may face difficult decisions. Advances in biotechnology offer the potential to reduce human suffering, but along with them come ethical issues that require conscientious deliberation.

Fetal Testing

Two types of testing can be done to look for genetic disorders in the fetus: screening tests and diagnostic tests. Screening tests are generally noninvasive and can be followed by diagnostic tests if screening results suggest that a genetic disorder might be present or if there are other risks, as when both parents are carriers.

Screening tests include imaging and blood tests. Imaging techniques allow a physician to examine a fetus directly for major anatomical abnormalities that might not show up in genetic tests. In the *ultrasound* technique, for example, reflected sound waves are used to produce an image of the fetus by a simple noninvasive procedure. An ultrasound is often accompanied by a blood test that looks for fetal proteins that might signal a genetic disorder.

Medical scientists have also developed methods for isolating and analyzing fetal DNA that has escaped into the maternal blood (*cell-free fetal DNA*), using various techniques. Cell-free fetal DNA tests and other blood tests are increasingly being used as noninvasive prenatal screening tests for certain chromosomal defects and disorders; a positive result indicates to the parents that further diagnostic testing should be considered.

One diagnostic test that can determine whether the developing fetus has a serious recessive disease is **amniocentesis**, which can be performed starting at the 15th or 16th week of pregnancy **(Figure 14.19a)**. In this procedure, a physician inserts a needle into the uterus and extracts about 10 mL of amniotic fluid, the liquid that bathes the fetus. Some genetic disorders can be detected from the presence of certain proteins or

hormones in the amniotic fluid itself. Tests for thousands of other disorders (such as cystic fibrosis) are performed on DNA from fetal cells sloughed off into the amniotic fluid using targeted DNA sequencing or a technique called a *DNA microarray* (see Figure 20.13) that tests for many alleles at the same time. A karyotype of these cultured cells can also confirm certain chromosomal defects that may not have been suggested by cell-free fetal DNA tests.

Figure 14.19

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Diagnostic fetal testing for genetic disorders.
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Diagnostic fetal testing for genetic disorders.

Most testing for genetic abnormalities is now DNA-based. In some cases biochemical tests are used to detect particular disorders, and karyotyping may be done to see whether the chromosomes of the fetus are normal in number and appearance.

Part a. Amniocentesis summarized in three steps.

- 1. A sample of amniotic fluid can be taken starting at the fifteenth week of pregnancy. Drawing of fetus in utero with placenta, fetus, uterus, cervix labeled. An ultrasound monitor is in place while the amniotic fluid is withdrawn using a needle.

- 2. Biochemical and genetic tests can be performed immediately on the amniotic fluid or later on the cultured cells. Arrows flow from withdrawn fluid in needle to fluid and fetal cells in a test tube, where after several hours biochemical and genetics tests are run. 3. Fetal cells must be cultured for several weeks to obtain sufficient numbers for karyotyping. Arrows flow from fetal cells in test tube to petri dish, where fetal cells are cultured for several weeks, after which a karyotype is made. Image of karyotype shows the fetus' 23 pairs of homologous chromosomes.

Part b. Chorionic villus sampling (CVS) described in two steps.

- 1. A sample of chorionic villus tissue can be taken as early as the tenth week of pregnancy. Drawing of fetus in utero with placenta, fetus, Chorionic villi, uterus and cervix labeled. Drawing shows suction tube inserted through cervix to collect fetal cells.

- 2. Karyotyping and biochemical and genetic tests can be performed on the fetal cells immediately, providing results within a day or so. Arrows flow to fetal cells in test tube.

After several hours, biochemical and genetic tests can be run, and a karyotype produced. Image of karyotype shows the fetus' 23 pairs of homologous chromosomes.

In an alternative diagnostic technique that is called **chorionic villus sampling (CVS)**, a physician inserts a narrow tube through the cervix into the uterus and suctions out a tiny sample of tissue from the placenta, the organ that transmits nutrients and fetal wastes between the fetus and the female parent **(Figure 14.19b)**. The cells of the chorionic villi of the placenta—the portion sampled—are derived from the fetus and have the same genotype and DNA sequence as the new individual. DNA from these cells can be analyzed for the presence of many genetic disorders. The main advantage of CVS is that it can be performed as early as the 10th or 11th week of pregnancy.

Ultrasound and isolation of fetal cells or DNA from maternal blood pose no known risk to either the female parent or the fetus, while amniocentesis and CVS can cause complications in a small percentage of cases (for both, between 1 in 1,000 and 1 in 500 cases). Amniocentesis or CVS for diagnostic testing is generally offered to females over age 35, due to their increased risk of bearing a child with Down syndrome or other chromosomal disorders, and may also be offered to younger females if there are risk factors or conditions suggested by family history. If the fetal tests reveal a serious disorder, the parents may face the difficult choice of either terminating the pregnancy or preparing to care for a child with a genetic disorder, one that might even be fatal. Parental and fetal screening for Tay-Sachs alleles done since 1980 has reduced the number of children born with this incurable disease by 90%.

Newborn Screening

Some genetic diseases and disorders can be detected at birth by simple biochemical tests that are routinely performed on blood from the baby's heel in most hospitals in the United States. One common screening program is for phenylketonuria (PKU), a recessively inherited disorder that occurs in about one out of every 10,000–15,000 births in the United States. Children with this disease cannot properly metabolize the amino acid phenylalanine. This compound and its by-product, phenylpyruvate, can accumulate to toxic levels in the blood, causing severe intellectual disability. However, if PKU is detected in the newborn, a specialized diet low in phenylalanine will usually allow for typical development. (Among many other substances, this diet excludes the artificial sweetener aspartame, which contains phenylalanine.)

The number of newborn genetic tests, many for metabolic disorders or enzyme deficiencies, is over 100 and ever-increasing; each state regulates which tests are used. If screening identifies a baby with a disease or disorder for which treatment can be started early on, that individual has improved chances for better life quality. Some diseases, like PKU, are treated by dietary changes and others by replacement of missing enzymes. Unfortunately, there are quite a few diseases for which no treatment is yet known.

Fetal and newborn screening for serious inherited diseases, tests for identifying carriers, and genetic counseling all rely on the Mendelian model of inheritance. We owe the “gene idea”—the concept of heritable factors transmitted according to simple rules of chance—to the elegant quantitative experiments of Gregor Mendel. The importance of his discoveries was overlooked by most biologists until the early 1900s, decades after he reported his findings. In the next chapter, you will learn how Mendel’s laws have their physical basis in the behavior of chromosomes during sexual life cycles and how the synthesis of Mendelian genetics and a chromosome theory of inheritance catalyzed progress in genetics.

Chapter 14 Review

A purple flower.

Summary of Key Concepts

Advanced placement edition

Why was Mendel’s choice of pea plants as an experimental organism so fortuitous in developing a model of inheritance? **(BIG IDEA 3)**

Concept 14.1: Mendel used the scientific approach to identify two laws of inheritance

- Gregor Mendel formulated a theory of inheritance based on experiments with garden peas, proposing that parents pass on to their offspring discrete genes that retain their identity through generations. This theory includes two “laws.”
- The ****__law of segregation__ states that genes have alternative forms, or ****__alleles.__ In a diploid organism, the two alleles of a gene segregate (separate) during meiosis and gamete formation; each sperm or egg carries only one allele of each pair. This law explains the 3:1 ratio of F2 ****__phenotypes__ observed when ****__monohybrids__ cross-or self-pollinate. Each organism inherits one allele for each gene from each parent. In ****__heterozygotes__, the two alleles are different: Expression of the ****__dominant allele__ masks the phenotypic effect of the ****__recessive allele__. ****__Homozygotes__ have identical alleles of a given gene and are therefore ****__true-breeding__.

- The ****law of independent assortment** states that the pair of alleles for a given gene segregates into gametes independently of the pair of alleles for any other gene. In a cross between ****dihybrids** (individuals heterozygous for two genes), the offspring have four phenotypes in a 9:3:3:1 ratio.

When Mendel did crosses of true-breeding purple- and white-flowered pea plants, the white-flowered trait disappeared from the F_1 *generation but reappeared in the* F_2 *generation. Use genetic terms to explain why that happened.*

Concept 14.2: Probability laws govern Mendelian inheritance

Image of coin toss representing segregation of alleles into sperm from genotype uppercase R lowercase r. One-half sperm are upper case R (heads); one-half sperm are lowercase r (tails).

- The ****multiplication rule** states that the probability of two or more events occurring together (a statement using *and*) is equal to the product of the individual probabilities of the independent single events. The ****addition rule** states that the probability of an event that can occur in two or more independent, mutually exclusive ways (using *or*) is the sum of the individual probabilities.
- The rules of probability can be used to solve complex genetics problems. A dihybrid or other multicharacter cross is equivalent to two or more independent monohybrid crosses occurring simultaneously. In calculating the chances of the various offspring genotypes from such crosses, each character is first considered separately and then the individual probabilities are multiplied.

DRAW IT *Redraw the Punnett square on the right side of* ***Figure 14.8*** *as two smaller monohybrid Punnett squares, one for each gene. Below each square, list the fractions of each phenotype produced. Use the rule of multiplication to compute the overall fraction of each of the possible dihybrid phenotypes. What is the phenotypic ratio?*

Concept 14.3: Inheritance patterns are often more complex than predicted by simple Mendelian genetics

- Extensions of Mendelian genetics for a single gene:

Example images show, 1, purple flower color, 2, Red, pink, white flower color, 3, Red blood cell with both type A and B carbohydrates.

Parts are as follows.

1. Complete dominance of one allele: P P both uppercase = purple flowers. Uppercase P lowercase p = purple flowers.
2. Incomplete dominance of either allele: C superscript R, C superscript R = red flowers. C superscript R, C superscript W= pink flowers. C superscript W, C superscript W = white flowers.
3. Codominance: Red blood cell with both A and B (circles and triangles) carbohydrates on the surface in an I superscript A, I superscript B individual.

- Extensions of Mendelian genetics for two or more genes:

Epistasis: Example: see Figure 14.12. Polygenic inheritance: Example: see Figure 14.13.

- The expression of a genotype can be affected by environmental influences, resulting in a range of phenotypes. Polygenic characters that are also influenced by the environment are called ****__multifactorial__ characters.
- An organism's overall phenotype reflects its overall genotype and unique environmental history. Even in more complex inheritance patterns, Mendel's fundamental laws still apply.

Which genetic relationships listed in the first column of the two tables above are demonstrated by the inheritance pattern of the ABO blood group alleles? For each genetic relationship, explain why this inheritance pattern is or is not an example.

Concept 14.4: Many human traits follow Mendelian patterns of inheritance

- Analysis of family ****__pedigrees__ can be used to deduce the possible genotypes of individuals and make predictions about future offspring. Such predictions are statistical probabilities rather than certainties.

Pedigree showing three generations of a family with respect to widow's peak. Images of two sisters shown in the third generation; one with widow's peak and one without.

Generations are as follows.

First generation (grandparents): Pedigree drawn with symbols as follows from left to right: First grandparents: Shaded Square (uppercase W lowercase w) with line drawn to circle (w w, both lowercase). Space and then second grandparents: square (w w both lowercase) with line drawn to shaded circle (uppercase W lower case w).

Second generation (parents, uncles and aunts): Children of first grandparents: Shaded circle (uppercase W lowercase w), square (w w both lowercase), square (w w both lowercase), shaded square (uppercase W

lowercase w). This individual has a line drawn to the first child of the second grandparents, a shaded circle (uppercase W lowercase w). Next is a circle (w w both lowercase).

Third generation: children of the couple in the second generation: Shaded circle (W W both uppercase or uppercase W lowercase w). This individual is shown next to a photograph of a woman with Widow's peak. Next is a circle (w w both lowercase). This individual is shown next to a photograph of a woman with no widow's peak.

Is a widow's peak a dominant or recessive trait? Tips for pedigree analysis: Notice in the third generation that the second-born daughter lacks a widow's peak, although both of her parents had the trait. Such a pattern indicates that the trait is due to a dominant allele. If it were due to a recessive allele, and both parents had the recessive phenotype (straight hairline), all of their offspring would also have the recessive phenotype.

- Many genetic disorders are inherited as simple recessive traits. Most affected (homozygous recessive) individuals are children of phenotypically normal, heterozygous ****__carriers__.
- The sickle-cell allele has probably persisted for evolutionary reasons: Homozygotes have **sickle-cell disease** ****, but heterozygotes have an advantage because one copy of the sickle-cell allele reduces both the frequency and severity of malaria attacks.

Homozygote with sickle-cell disease. Cascade of events in a homozygous individual that leads to sickled red blood cells.

Homozygote with sickle-cell disease: Weakness, anemia, pain and fever, organ damage. Pair of homologous chromosomes, each having a sickle-cell allele, flows to both sickle-cell hemoglobin proteins, flows with low O_2 to part of a fiber of sickle-cell hemoglobin proteins. Long fibers cause red blood cells to be sickle-shaped and causes sickle-cell disease.

- Lethal dominant alleles are eliminated from the population if affected people die before reproducing. Nonlethal dominant alleles and lethal ones that strike relatively late in life can be inherited in a Mendelian way.
- Many human diseases are multifactorial—that is, they have both genetic and environmental components and do not follow simple Mendelian inheritance patterns.
- Using family histories, genetic counselors help couples determine the probability that their children will have genetic disorders. Genetic testing of prospective parents can reveal whether they are carriers of recessive alleles associated with specific disorders. Ultrasounds and blood tests can screen for certain disorders in a fetus. ****__Amniocentesis__ and ****__chorionic villus sampling__

can diagnose whether a suspected genetic disorder is present in a fetus. Other genetic tests can be performed after birth.

Both members of a couple know that they are carriers of the cystic fibrosis allele. None of their three children have cystic fibrosis, but any one of them might be a carrier. The couple would like to have a fourth child but are worried that he or she would very likely have the disease, since the first three do not. What would you tell the couple? Would it remove some uncertainty from their prediction if they could find out whether the three children are carriers?

Tips for Genetics Problems

- Write down symbols for the alleles. (These may be given in the problem.) When represented by single letters, the dominant allele is uppercase and the recessive is lowercase.
- Write down the possible genotypes, as determined by the phenotype.
- If the phenotype is that of the dominant trait (for example, purple flowers), then the genotype is either homozygous dominant or heterozygous (PP or Pp in this example).
- If the phenotype is that of the recessive trait, the genotype must be homozygous recessive (for example, pp).
- If the problem says “true-breeding,” the genotype is homozygous.
- Determine what the problem is asking for. If asked to do a cross, write it out in the form [genotype]×[genotype], using the alleles you’ve decided on.
- To figure out the outcome of a cross, set up a Punnett square.
- Put the gametes of one parent at the top and those of the other on the left. To determine the allele(s) in each gamete for a given genotype, set up a systematic way to list all the possibilities. (Remember, each gamete has one allele of each gene.) Note that there are 2n possible types of gametes, where n is the number of gene loci that are heterozygous. For example, an individual with genotype AaBbCc would produce 23=8 types of gametes. Write the genotypes of the gametes in circles above the columns and to the left of the rows.
- Fill in the Punnett square as if each possible sperm were fertilizing each possible egg, making all of the possible offspring. In a cross of AaBbCc×AaBbCc, for example, the Punnett square would have 8 columns and 8 rows, so there are 64 different offspring; you would know the genotype of each and thus the phenotype. Count genotypes and phenotypes to obtain the genotypic and phenotypic ratios. Because the Punnett square is so large, this method is not the most efficient. See tip 5.

- You can use the rules of probability if a Punnett square would be too big. (For example, see the question at the end of the summary for Concept 14.2 and question 7 below.) You can consider each gene separately (see the section Solving Complex Genetics Problems with the Rules of Probability in Concept 14.2).
- If the problem gives you the phenotypic ratios of offspring but not the genotypes of the parents in a given cross, the phenotypes can help you deduce the parents' unknown genotypes.
- For example, if 12 of the offspring have the recessive phenotype and 12 the dominant, you know that the cross was between a heterozygote and a homozygous recessive.
- If the ratio is 3:1, the cross was between two heterozygotes.
- If two genes are involved and you see a 9:3:3:1 ratio in the offspring, you know that each parent is heterozygous for both genes. Caution: Don't assume that the reported numbers will exactly equal the predicted ratios. For example, if there are 13 offspring with the dominant trait and 11 with the recessive, assume that the ratio is one dominant to one recessive.
- For pedigree problems, use the tips in Figure 14.15 and below to determine what kind of trait is involved.
- If parents without the trait have offspring with the trait, the trait must be recessive and the parents both carriers.
- If the trait is seen in every generation, it is most likely dominant (see the next possibility, though).
- If both parents have the trait, then in order for it to be recessive, all offspring must show the trait.
- To determine the likely genotype of a certain individual in a pedigree, first label the genotypes of all the family members you can. Even if some of the genotypes are incomplete, label what you do know. For example, if an individual has the dominant phenotype, the genotype must be AA or Aa; you can write this as A–. Try different possibilities to see which fits the results. Use the rules of probability to calculate the probability of each possible genotype being the correct one.